The Hyperbolic Sparsemax Pump:
A Rational, Parity-Safe Executable Kernel for the SHD–CCP Lattice

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Abstract

We construct a deterministic transition kernel (the *Hyperbolic Sparsemax Kernel*) on the 64-voxel toroidal lattice $V = (\mathbb{Z}/4\mathbb{Z})^3$ underlying the Symbolic High-Dimensional Context Compression Protocol (SHD–CCP), and prove that it is rational, row-stochastic, exactly sparse, and free of transcendental operations. Each voxel carries a rational unit quaternion whose scalar part is confined to a bounded annulus. A square-root-free Lorentz lift sends each spinor to an exactly rational point of the hyperboloid $\mathbb{H}^3$, on which a rational hyperbolic quadrance $Q_h = \langle \tilde{q}_i, \tilde{q}_j \rangle_{\mathbb{L}}^2 - 1 = \sinh^2 d_{\mathbb{H}}$ acts as a symmetric, nonnegative, point-separating metric kernel. Logits formed from Q_h, a bounded inverse temperature, and a torsional resistance term are projected onto the probability simplex by *sparsemax*. Because this projection uses only field arithmetic, comparison, and positive-part thresholding, the entire executable pipeline is closed in $\mathbb{Q}$; the computation is therefore bit-reproducible across hardware and the discrete SHD–CCP parity invariant is protected. Equipping the sparsemax support with a symmetric rational base walk and a rational reverse-valve yields an executable kernel $\widehat{K}$ that remains rational and row-stochastic while breaking detailed balance relative to the reversible base measure, with a nonnegative (and, under a mild support condition, strictly positive) base-referenced forward current. Replacing the rational valve by an exponential valve recovers the analytic softmax companion (the Torsional Trefoil Markov Pump). All stated results are proved; we close by delimiting the scope of the current-positivity statement and the role of the topological vocabulary.

1 Introduction

The analytic companion to this work (the Torsional Trefoil Markov Pump) provides a continuous mechanism that breaks detailed balance and advances a longitudinal binding scalar through an asymmetric transition kernel. Realising such dynamics on discrete, dynamically reconfigurable hardware is delicate: transcendental primitives (exponentials, logarithms, square roots, inverse hyperbolic functions) are evaluated in floating point, and the resulting rounding drift can corrupt the discrete invariants the protocol relies on to avoid gridlock.

This paper isolates the part of the construction that can be executed *exactly*. We reserve the words *softmax* and *exponential attenuation* for the analytic limit, and on the executable lattice we use instead a *sparsemax* rectifier together with a rational attenuation valve. The key observation is structural rather than numerical: sparsemax is a Euclidean projection onto the probability simplex, so it is computed with arithmetic, comparison, and thresholding alone and never leaves the field $\mathbb{Q}$. Combined with a square-root-free lift to the hyperboloid, this keeps every executable quantity rational, which is exactly what makes the pipeline deterministic and reproducible across devices.

We carefully separate the spinor that encodes local orientation from the point that carries the hyperbolic score: orientation lives on the quaternion sphere S^3, while scoring is performed after a Lorentz lift to the hyperboloid $\mathbb{H}^3$. With this separation in place, the remainder of the paper proves that the executable kernel is rational, row-stochastic, sparse, parity-safe, and capable of supporting a base-referenced forward current, and that its analytic completion reproduces the continuous pump.

2 Lattice, Spinors, and Standing Assumptions

The lattice. The ambient discrete space is the toroidal lattice

$$V = (\mathbb{Z}/4\mathbb{Z})^3, \qquad |V| = 64,$$

partitioned into a core of 8 voxels and a mantle of 56 voxels. A voxel is indexed by a triple of residues modulo 4. In the SHD–CCP interpretation the three axes correspond to the three arms of the trefoil and the four coordinate positions to four phases; torsional chirality and phase-lifting resolve the six signed face-neighbor directions into a 12-channel coupling space, in which several channels may project onto the same lattice neighbor while remaining distinct before row aggregation. This labeling fixes the chirality and resistance conventions used below; as recorded in the scope remarks of Section 13, the theorems themselves do not depend on it.

Spinors and the lift. Each voxel $i \in V$ carries a rational spinor

$$q_i = (w_i, \mathbf{m}_i) \in S^3 \cap \mathbb{Q}^4, \qquad \mathbf{m}_i = (x_i, y_i, z_i), \qquad w_i^2 + \|\mathbf{m}_i\|^2 = 1.$$

We write the Minkowski bilinear form on $\mathbb{R}^4$ as

$$\langle a,\, b\rangle_{\mathbb{L}} = a_0 b_0 - \mathbf{a} \cdot \mathbf{b}.$$

Standing assumptions.

Assumption A1 (Rational annular spinors)**.** Every spinor is a rational unit quaternion with scalar part bounded away from both extremes,

$$0 < w_{\min} \le w_i \le w_{\max} < 1.$$

Assumption A2 (Rational hardware constraints)**.** The torsional resistance is $A_{ij} = A_0\, s_{ij}\, \chi_{ij}$ with $A_0, s_{ij} \in \mathbb{Q}_{\ge 0}$, attenuation strength $\lambda \in \mathbb{Q}_{>0}$, and chirality indicator $\chi_{ij} \in \{0, 1\}$ (0 on forward edges, 1 on reverse edges).

Assumption A3 (Candidate neighborhood)**.** For each voxel i, the finite candidate set $\mathcal{N}_i$ consists of the self-loop $i \to i$ together with the face-neighbor transitions resolved through the 12 torsional channel labels.

Assumption A4 (Octet residual quarantine)**.** No transcendental function is evaluated in the executable 64-state transition path. Irrational residuals are recorded only in a separate parity buffer and are never fed back into the rational kernel before rationalization.

Our objective is to define a transition row $S_{i\cdot}$ over $\mathcal{N}_i$, using only rational operations, such that every entry is a nonnegative rational, the row sums to one, most entries are exactly zero, and no exponential, root, or trigonometric evaluation is required.

3 The Rational Lorentz Lift

Definition 1 (Lorentz lift)**.** For $q_i = (w_i, \mathbf{m}_i)$ with $w_i > 0$ define

$$\Phi(q_i) = \tilde{q}_i = \frac{1}{w_i}(1, \mathbf{m}_i) = \left(\frac{1}{w_i}, \frac{x_i}{w_i}, \frac{y_i}{w_i}, \frac{z_i}{w_i}\right).$$

Theorem 2 (Closed rational lift)**.** *Under Assumption* A1*, for every i the lift $\tilde{q}_i$ has rational coordinates and lies on the positive sheet of the hyperboloid*

$$\mathbb{H}^3 = \{\tilde{q} : \langle \tilde{q}, \tilde{q} \rangle_{\mathbb{L}} = 1,\ \tilde{q}_0 > 0\}.$$

Proof. Compute the Minkowski self-product:

$$\langle \tilde{q}_i, \tilde{q}_i \rangle_{\mathbb{L}} = \left(\tfrac{1}{w_i}\right)^2 - \left\|\tfrac{\mathbf{m}_i}{w_i}\right\|^2 = \frac{1 - \|\mathbf{m}_i\|^2}{w_i^2}.$$

Since $q_i \in S^3$ we have $w_i^2 + \|\mathbf{m}_i\|^2 = 1$, hence $1 - \|\mathbf{m}_i\|^2 = w_i^2$ and

$$\langle \tilde{q}_i, \tilde{q}_i \rangle_{\mathbb{L}} = \frac{w_i^2}{w_i^2} = 1.$$

Moreover $\tilde{q}_{i,0} = 1/w_i > 0$, so $\tilde{q}_i \in \mathbb{H}^3$. All coordinates are quotients of rationals with nonzero rational denominator w_i, hence rational. □

Remark 3. The lift exploits the identity $w_i = \sqrt{1 - \|\mathbf{m}_i\|^2}$, which holds automatically for a unit quaternion. Storing the scalar part explicitly therefore removes the square root that the standard ball-to-hyperboloid map would require, and is what keeps Φ inside $\mathbb{Q}$. The orientation spinor $q_i \in S^3$ and its hyperbolic score point $\tilde{q}_i \in \mathbb{H}^3$ are distinct objects related by Φ.

4 Hyperbolic Quadrance

Definition 4 (Hyperbolic quadrance)**.** For lifted points $\tilde{q}_i, \tilde{q}_j$ define

$$Q_h(i, j) = \langle \tilde{q}_i, \tilde{q}_j \rangle_{\mathbb{L}}^2 - 1.$$

This is a rational surrogate for the hyperbolic distance $d_{\mathbb{H}}(i, j) = \operatorname{arcosh} \langle \tilde{q}_i, \tilde{q}_j \rangle_{\mathbb{L}}$, whose evaluation would introduce a transcendental function. Since $\langle \tilde{q}_i, \tilde{q}_j \rangle_{\mathbb{L}} = \cosh d_{\mathbb{H}}$, one has $Q_h = \cosh^2 d_{\mathbb{H}} - 1 = \sinh^2 d_{\mathbb{H}}$, a monotone, rational function of distance.

Theorem 5 (Quadrance is a rational metric kernel)**.** *For all i, j,*

$$Q_h(i, j) \in \mathbb{Q}_{\geq 0}, \qquad Q_h(i, j) = Q_h(j, i),$$

$$Q_h(i, j) = 0 \iff q_i = q_j.$$

Proof. Rationality and symmetry. From Definition 1,

$$\langle \tilde{q}_i, \tilde{q}_j \rangle_{\mathbb{L}} = \frac{1}{w_i w_j} - \frac{\mathbf{m}_i \cdot \mathbf{m}_j}{w_i w_j} = \frac{1 - \mathbf{m}_i \cdot \mathbf{m}_j}{w_i w_j} \in \mathbb{Q},$$

so $Q_h(i,j) = \big((1 - \mathbf{m}_i \cdot \mathbf{m}_j)/(w_i w_j)\big)^2 - 1$ is rational, and it is symmetric because $\mathbf{m}_i \cdot \mathbf{m}_j = \mathbf{m}_j \cdot \mathbf{m}_i$.

Nonnegativity. As unit vectors in $\mathbb{R}^4$, the Cauchy–Schwarz inequality gives $q_i \cdot q_j \leq 1$. Writing $q_i \cdot q_j = w_i w_j + \mathbf{m}_i \cdot \mathbf{m}_j$ yields $w_i w_j + \mathbf{m}_i \cdot \mathbf{m}_j \leq 1$, i.e.

$$1 - \mathbf{m}_i \cdot \mathbf{m}_j \;\geq\; w_i w_j > 0.$$

Dividing by $w_i w_j > 0$ gives $\langle \tilde{q}_i, \tilde{q}_j \rangle_{\mathbb{L}} \geq 1$, hence

$$Q_h(i,j) = \langle \tilde{q}_i, \tilde{q}_j \rangle_{\mathbb{L}}^2 - 1 \geq 0.$$

Point separation. Equality $\langle \tilde{q}_i, \tilde{q}_j \rangle_{\mathbb{L}} = 1$ is equivalent to $q_i \cdot q_j = 1$, which for unit vectors holds iff $q_i = q_j$. The lift is injective: from $\tilde{q}_i$ one recovers $w_i = 1/\tilde{q}_{i,0}$ and then $\mathbf{m}_i = w_i\,(\tilde{q}_{i,1}, \tilde{q}_{i,2}, \tilde{q}_{i,3})$. Therefore $Q_h(i,j) = 0 \iff \langle \tilde{q}_i, \tilde{q}_j \rangle_{\mathbb{L}} = 1 \iff q_i = q_j$. □

5 Rational Hyperbolic Logits

Definition 6 (Inverse temperature and executable logit)**.** For each i set the bounded inverse temperature

$$\beta_i = \frac{1 - w_i}{w_i} \in \mathbb{Q}_{>0},$$

and for each candidate $j \in \mathcal{N}_i$ define the executable logit

$$z_{i \to j} = -\beta_i\, Q_h(i,j) - \lambda\, A_{ij}.$$

Lemma 7 (Rational logits)**.** *Under Assumptions* A1–A2*, $z_{i \to j} \in \mathbb{Q}$ for every $j \in \mathcal{N}_i$.*

Proof. By Theorem 5, $Q_h(i,j) \in \mathbb{Q}$; by hypothesis $\beta_i, \lambda, A_{ij} \in \mathbb{Q}$. The field $\mathbb{Q}$ is closed under addition and multiplication, so $-\beta_i Q_h(i,j) - \lambda A_{ij} \in \mathbb{Q}$. □

6 Sparsemax Projection

Definition 8 (Sparsemax row)**.** Let $\Delta_i = \{p \in \mathbb{R}^{|\mathcal{N}_i|} : p_j \geq 0,\ \sum_{j \in \mathcal{N}_i} p_j = 1\}$ be the probability simplex on $\mathcal{N}_i$. The sparsemax row is the Euclidean projection of the logit vector $z_{i\cdot}$ onto Δ_i:

$$S_{i\cdot} = \operatorname{sparsemax}(z_{i\cdot}) = \operatorname*{arg\,min}_{p \in \Delta_i} \; \|p - z_{i\cdot}\|_2^2 .$$

Equivalently, sparsemax is the $\alpha = 2$ member of the entmax family, whose $\alpha = 1$ limit is softmax.

Theorem 9 (Closed form)**.** *The minimizer in Definition* 8 *is unique and given by*

$$S_{ij} = [z_{i\to j} - \tau_i]_+, \qquad [x]_+ = \max(x, 0),$$

where τ_i is the unique value with $\sum_{j\in\mathcal{N}_i} [z_{i\to j} - \tau_i]_+ = 1$. If $A_i = \{j : z_{i\to j} > \tau_i\}$ denotes the active set, then

$$\tau_i = \frac{\sum_{j\in A_i} z_{i\to j} - 1}{|A_i|}.$$

Proof. Minimize $\frac{1}{2}\|p - z\|_2^2$ over Δ_i via the Lagrangian

$$\mathcal{L}(p, \tau, \mu) = \tfrac{1}{2}\sum_j (p_j - z_j)^2 + \tau\Big(\sum_j p_j - 1\Big) - \sum_j \mu_j p_j, \qquad \mu_j \geq 0.$$

Stationarity gives $p_j - z_j + \tau - \mu_j = 0$, i.e. $p_j = z_j - \tau + \mu_j$. Complementary slackness $\mu_j p_j = 0$ forces $p_j = z_j - \tau$ when $p_j > 0$ (so $\mu_j = 0$), and $z_j - \tau \leq 0$ when $p_j = 0$. Both cases are summarized by $p_j = [z_j - \tau]_+$. The multiplier τ is fixed by feasibility $\sum_j [z_j - \tau]_+ = 1$; on the active set $A_i = \{j : z_j > \tau\}$ this reads $\sum_{j\in A_i}(z_j - \tau) = 1$, whence $\tau = \big(\sum_{j\in A_i} z_j - 1\big)/|A_i|$. Strict convexity of the squared Euclidean norm on the convex set Δ_i gives uniqueness. □

7 The Hyperbolic Sparsemax Kernel

Definition 10 (Kernel)**.** The *Hyperbolic Sparsemax Kernel* is the row operator

$$S_{ij} = \mathrm{sparsemax}_j\big(-\beta_i Q_h(i, j) - \lambda A_{ij}\big) = [z_{i\to j} - \tau_i]_+ .$$

Theorem 11 (Executable validity)**.** *Under Assumptions* A1–A4 *the row $S_{i\cdot}$ satisfies:*

(i) (Rationality) $S_{ij} \in \mathbb{Q}$ *for all j;*

(ii) (Probability row) $S_{ij} \geq 0$ *and* $\sum_{j\in\mathcal{N}_i} S_{ij} = 1$*;*

(iii) (Exact sparsity) $S_{ij} = 0$ *whenever* $z_{i\to j} \leq \tau_i$*;*

(iv) (Nonempty support) *the active set $A_i = \{j : S_{ij} > 0\}$ is nonempty, and the self-loop is always among the maximizers.*

Proof. (i) By Lemma 7 the logits are rational. The active set A_i is determined by comparisons of rationals, which are exact; the threshold $\tau_i = (\sum_{j\in A_i} z_{i\to j} - 1)/|A_i|$ has rational numerator and positive-integer denominator, so $\tau_i \in \mathbb{Q}$. Each $S_{ij} = [z_{i\to j} - \tau_i]_+$ is then either a rational or 0, hence rational.

(ii) Nonnegativity is immediate from the positive part, and τ_i is defined precisely so that $\sum_j [z_{i\to j} - \tau_i]_+ = 1$.

(iii) If $z_{i\to j} \leq \tau_i$ then $z_{i\to j} - \tau_i \leq 0$, so $[z_{i\to j} - \tau_i]_+ = 0$; inactive candidates receive exactly zero mass, not a small positive value.

(iv) If every S_{ij} were 0 the row sum would be 0, contradicting (ii); hence $A_i \neq \varnothing$. For the self-loop, Theorem 5 gives $Q_h(i, i) = 0$, and with $A_{ii} = 0$ we obtain $z_{i\to i} = 0$. For any other candidate $Q_h(i, j) \geq 0$ and $A_{ij} \geq 0$, so $z_{i\to j} = -\beta_i Q_h(i, j) - \lambda A_{ij} \leq 0 \leq z_{i\to i}$. Thus the self-logit is maximal and the self-loop lies among the maximizers, preventing support collapse. □

8 Symmetric Base Walk and Rational Reverse-Valve

To turn the sparse kernel into a pump we add a rational asymmetry layer on its support.

Definition 12 (Reverse-valve)**.** For $A_{ij} \geq 0$ define

$$\rho_{ij} = \frac{1}{1 + \lambda A_{ij}} \in \mathbb{Q}, \qquad 0 < \rho_{ij} \leq 1,$$

with $\rho_{ij} < 1$ exactly when $A_{ij} > 0$.

Definition 13 (Symmetric base walk)**.** Let the positive sparsemax support determine the admissible undirected edges, and choose symmetric rational weights $W_{ij} = W_{ji} \in \mathbb{Q}_{\geq 0}$ on them. Set

$$d_i = \sum_j W_{ij}, \qquad D = \sum_i d_i, \qquad \pi_i^0 = \frac{d_i}{D}, \qquad P_{ij}^0 = \frac{W_{ij}}{d_i}.$$

Proposition 14 (Reversibility of the base walk)**.** *P^0 is reversible with respect to π^0.*

Proof. $\pi_i^0 P_{ij}^0 = \dfrac{d_i}{D}\dfrac{W_{ij}}{d_i} = \dfrac{W_{ij}}{D}$ and likewise $\pi_j^0 P_{ji}^0 = \dfrac{W_{ji}}{D}$; since $W_{ij} = W_{ji}$, detailed balance $\pi_i^0 P_{ij}^0 = \pi_j^0 P_{ji}^0$ holds. □

Definition 15 (Valve-modified pump kernel)**.** Partition the supported oriented edges into a forward set E^+ (edges following forward chirality) and the reverse set $E^- = \{(j,i) : (i,j) \in E^+\}$. Define

$$\widehat{K}_{ij} = \begin{cases} P_{ij}^0, & (i,j) \in E^+, \\ P_{ij}^0\, \rho_{ij}, & (i,j) \in E^-, \\ P_{ii}^0 + \displaystyle\sum_{k:(i,k)\in E^-} P_{ik}^0 \big(1 - \rho_{ik}\big), & j = i. \end{cases}$$

Reverse mass is attenuated and the rejected mass is returned to the diagonal.

Theorem 16 ($\widehat{K}$ is rational and row-stochastic)**.** *For all i, j, $\widehat{K}_{ij} \in \mathbb{Q}$, and for every i, $\sum_j \widehat{K}_{ij} = 1$.*

Proof. All of $W_{ij}, d_i, D, P_{ij}^0, \rho_{ij}$ are rational, so each entry of $\widehat{K}$ is rational. Fix a row i. Every supported off-diagonal target is either forward or reverse, so starting from $\sum_j P_{ij}^0 = 1$ the only changes are, for each reverse edge (i,k), a reduction of the off-diagonal mass from P_{ik}^0 to $P_{ik}^0 \rho_{ik}$, i.e. a removal of $P_{ik}^0(1-\rho_{ik})$, which is added back to the diagonal. The reverse contributions recombine as $P_{ik}^0 \rho_{ik} + P_{ik}^0(1 - \rho_{ik}) = P_{ik}^0$, so

$$\sum_j \widehat{K}_{ij} = \sum_{j:(i,j)\in E^+} P_{ij}^0 + \sum_{k:(i,k)\in E^-} P_{ik}^0 + P_{ii}^0 = \sum_j P_{ij}^0 = 1.$$ □

9 Broken Detailed Balance and Forward Current

Theorem 17 (Base-referenced forward current)**.** *For a forward edge $(i,j) \in E^+$ define the base-referenced current*

$$J^0_{ij} = \pi^0_i \widehat{K}_{ij} - \pi^0_j \widehat{K}_{ji}.$$

Then

$$J^0_{ij} = \frac{W_{ij}}{D}\big(1 - \rho_{ji}\big) \;\geq\; 0,$$

with $J^0_{ij} > 0$ whenever $W_{ij} > 0$ and $\rho_{ji} < 1$. Consequently $\widehat{K}$ is not reversible with respect to π^0, and the total forward current satisfies

$$J_0 = \sum_{(i,j)\in E^+} J^0_{ij} = \sum_{(i,j)\in E^+} \frac{W_{ij}}{D}\big(1 - \rho_{ji}\big) \;\geq\; 0,$$

with $J_0 > 0$ as soon as at least one supported forward edge has $W_{ij} > 0$ and $\rho_{ji} < 1$.

Proof. For $(i,j) \in E^+$ we have $\widehat{K}_{ij} = P^0_{ij}$, while the reverse transition is $\widehat{K}_{ji} = P^0_{ji}\rho_{ji}$ because $(j,i) \in E^-$. Hence

$$J^0_{ij} = \pi^0_i P^0_{ij} - \pi^0_j P^0_{ji}\rho_{ji}.$$

By Proposition 14, $\pi^0_i P^0_{ij} = \pi^0_j P^0_{ji} = W_{ij}/D$, so

$$J^0_{ij} = \frac{W_{ij}}{D} - \frac{W_{ij}}{D}\rho_{ji} = \frac{W_{ij}}{D}\big(1 - \rho_{ji}\big).$$

Since $W_{ij} \geq 0$ and $0 < \rho_{ji} \leq 1$, we get $J^0_{ij} \geq 0$, strictly positive when $W_{ij} > 0$ and $\rho_{ji} < 1$. A single edge with $J^0_{ij} \neq 0$ already violates detailed balance with respect to π^0. Summing over E^+ gives the stated total current. □

10 Determinism and Parity Safety

Theorem 18 (Closure in $\mathbb{Q}$)**.** *Under Assumptions* A1–A4*, every executable quantity of the Hyperbolic Sparsemax Pump lies in $\mathbb{Q}$: the spinors q_i, the lifts $\tilde{q}_i$, the quadrances $Q_h(i,j)$, the logits $z_{i\to j}$, the thresholds τ_i, the sparsemax entries S_{ij}, the base weights W_{ij}, the valve factors ρ_{ij}, and the pump entries $\widehat{K}_{ij}$.*

Proof. Theorems 2, 5, Lemma 7, and Theorems 11, 16 establish rationality of each listed quantity. Every step uses only field operations, comparisons of rationals, and the positive part $\max(\cdot, 0)$, each of which maps $\mathbb{Q}$ to $\mathbb{Q}$. By Assumption A4 no transcendental value enters the 64-state path. Hence the pipeline is closed in $\mathbb{Q}$. □

Corollary 19 (Parity safety)**.** *If rationals are stored in canonical reduced form, then two nodes sharing a seed compute bitwise-identical rows $S_{i\cdot}$ and pump rows $\widehat{K}_{i\cdot}$. No floating-point drift can therefore alter a discrete parity invariant such as*

$$\text{torusWinding}(\textit{mantle}) \bmod 7 = \textit{core}.$$

Proof. Canonical reduced fractions are unique and hardware-agnostic, so by Theorem 18 the entire computation is exact and reproducible; the discrete modular relation is a function of exactly computed integers and is preserved. □

11 Annular (Derrick) Stability

Proposition 20 (Bounded inverse temperature)**.** *Under Assumption* A1*, the inverse temperature is uniformly bounded:*

$$0 < \frac{1 - w_{\max}}{w_{\max}} \leq \beta_i \leq \frac{1 - w_{\min}}{w_{\min}} < \infty.$$

Consequently the kernel avoids the delta-collapse limit $\beta_i \to \infty$ (forced as $w_i \to 0^+$) and the uniform-diffusion limit $\beta_i \to 0$ (forced as $w_i \to 1^-$).

Proof. The map $w \mapsto (1-w)/w$ is strictly decreasing on $(0,1)$ and continuous on the closed annulus $[w_{\min}, w_{\max}]$, so it attains the displayed finite, positive rational bounds. Neither degenerate limit is reachable from within the annulus. □

Ties in the active set are separated either by a bounded rational perturbation of β_i (or of the active-set threshold) when the tied alternatives differ in quadrance or resistance, or otherwise by a deterministic canonical rule, e.g. lexicographic voxel–channel ordering. The kernel therefore remains stable and deterministic.

12 Synthesis

Theorem 21 (Hyperbolic Sparsemax Pump)**.** *Let*

$$V = (\mathbb{Z}/4\mathbb{Z})^3, \qquad q_i \in S^3 \cap \mathbb{Q}^4,$$

$$w_i \in [w_{\min}, w_{\max}] \subset (0,1),$$

$$A_{ij} = A_0 s_{ij} \chi_{ij} \in \mathbb{Q}_{\geq 0}, \qquad \lambda \in \mathbb{Q}_{>0}.$$

Then the construction

$$\tilde{q}_i = \frac{1}{w_i}(1, \mathbf{m}_i), \qquad Q_h(i,j) = \langle \tilde{q}_i, \tilde{q}_j \rangle_{\mathbb{L}}^2 - 1,$$

$$z_{i \to j} = -\beta_i Q_h(i,j) - \lambda A_{ij}, \qquad S_{ij} = [z_{i \to j} - \tau_i]_+$$

defines a rational, exactly sparse, row-stochastic executable kernel (Theorems 2–11). When equipped with a symmetric rational base walk P^0 and the rational reverse-valve $\rho_{ij} = 1/(1+\lambda A_{ij})$ that returns rejected reverse mass to the diagonal, the resulting kernel $\widehat{K}$ remains rational and row-stochastic (Theorem 16) while breaking detailed balance relative to π^0, with base-referenced forward current

$$J_0 = \sum_{(i,j) \in E^+} \left(\pi_i^0 \widehat{K}_{ij} - \pi_j^0 \widehat{K}_{ji} \right) \geq 0,$$

strictly positive whenever some supported forward edge has nonzero affinity and nontrivial reverse attenuation (*Theorem* 17)*. The whole executable pipeline is closed in* $\mathbb{Q}$ *and parity-safe* (*Theorem* 18*, Corollary* 19)*, and is annularly stable* (*Proposition* 20)*.*

Analytic companion. If the rational valve is replaced by the exponential valve $\rho_{ij} = e^{-\lambda A_{ij}}$ and sparsemax by the Gibbs/softmax row, the construction reproduces the analytic Torsional Trefoil Markov Pump, for which the forward current is strictly positive on every supported forward edge. The rational kernel above is the executable, parity-safe counterpart of that analytic limit.

13 Discussion and Scope

Remark 22 (On the current)*.* Theorem 17 establishes that $\widehat{K}$ is not reversible with respect to the base measure π^0 and that the induced *base-referenced* current is nonnegative, and strictly positive under the stated support condition. The stronger claim that $\widehat{K}$ sustains a nonzero stationary probability current under its own invariant measure $\hat{\pi}$ (satisfying $\hat{\pi}\widehat{K} = \hat{\pi}$) is not asserted here; computing $\hat{\pi}$ and its circulation is a natural next step and is what would justify the "pump" interpretation in the strongest sense.

Remark 23 (Relation to established results)*.* The construction reuses standard components: the sparsemax projection and its threshold characterization [1, 3], the entmax family that interpolates to softmax [2], the reversible random walk on a weighted graph [6], and the hyperboloid model of hyperbolic space [5]; the term *quadrance* follows rational trigonometry [4]. The specific contribution is their exact-rational realization: a square-root-free Lorentz lift together with a sparsemax (rather than softmax) rectifier keep every executable quantity in $\mathbb{Q}$, which is precisely what makes the pipeline bit-reproducible and the parity invariant safe.

Remark 24 (On the topological vocabulary)*.* The trefoil arms, torsional chirality, annular (Derrick) stability, and the modular parity invariant supply modeling motivation and fix the chirality and resistance conventions of Section 2. The theorems above hold for *any* rational annular spinor field on V and any symmetric rational base weight on the sparsemax support, and do not depend on this interpretation.

14 Conclusion

We have given a deterministic, strictly rational, hardware-executable transition kernel on the 64-voxel SHD–CCP lattice. By scoring with a square-root-free Lorentz lift and a rational hyperbolic quadrance, and by rectifying with sparsemax rather than softmax, the executable pipeline never leaves $\mathbb{Q}$: it is rational, row-stochastic, exactly sparse, bit-reproducible, and parity-safe. A symmetric base walk with a rational reverse-valve breaks detailed balance relative to the reversible base measure and produces a nonnegative, conditionally strictly positive base-referenced forward current, while the exponential limit recovers the analytic pump. The Hyperbolic Sparsemax Pump is thus the parity-safe, executable member of the family whose analytic completion is the Torsional Trefoil Markov Pump.

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